



A hierarchical single-pulse shock tube pyrolysis study of C₂–C₆ 1-alkenes

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ABSTRACT

A single-pulse shock tube study of the pyrolysis of 2% C₂–C₆ 1-alkenes is presented at 2 bar in the temperature range 900–1800 K in the current study. Reactant, intermediate and product species are obtained and quantified using gas chromatography-mass spectrometry (GC–MS) analysis. MS is used for species identification and a flame ionization detector is used for quantification. The experiments show the effect of carbon chain length on the production of smaller C₁–C₃ fragments. A new detailed kinetic mechanism, NUIGMech1.0, is used to simulate the data and the predictions for the major species are satisfactory. The improvement in predictions of the current mechanism, which includes C₆ and C₇ species, is substantial compared to AramcoMech3.0. Kinetic analyses are conducted with the current mechanism to identify the formation and consumption pathways of the quantified species. The experimental data are expected to contribute to a database for the validation of mechanisms at pyrolytic conditions. Additionally, the mechanism aims to provide a fundamental understanding of the pyrolytic chemistry of olefins.

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1. Introduction

Chemical kinetic models are needed to simulate fuel chemistry in combustion devices and to aid in the development of new designs to reduce harmful emissions and maximize combustor performance. However, these models need to be validated using fundamental experimental data to ensure that they are accurate in their predictions. Shock tubes, rapid compression machines, plug flow reactors, jet-stirred reactors and constant volume bombs are some of the widely used experimental facilities for such validation data. Even though laminar flame speeds and ignition delay time measurements provide significant validation data of a fuel's reactivity, the information provided by the measurement of the concentration of reactants, intermediates and product species formed during pyrolytic or oxidative processes are extremely valuable as they provide evidence of the chemical changes occurring and place a greater predictive burden on a chemistry model.

Shock tube speciation techniques can broadly be classified into two categories: (a) post-shock sampled and (b) time-resolved in-situ species measurements. A single-pulse shock tube (SPST) is a well-established reactor for post-shock sampling and analysis. The first design is attributed to Glick et al. [1] and other designs have been implemented by Tsang [2], Hidaka et al. [3], and Tranter et al. [4] among others. Given that a SPST has a quenching rate of $\sim 6 \times 10^6 \text{ K s}^{-1}$ after the residence time (classically defined as the time between the arrival of the reflected shock to when the pressure has fallen by 20% [5]), it can be assumed that the reaction is frozen due to rapid quenching.

Olefins are important intermediates in the thermal decomposition and oxidation of hydrocarbons. Pyrolytic studies of these fuels are particularly important in understanding combustion because, at higher combustion temperatures, fuels undergo pyrolytic reactions as an initiation step. However, pyrolysis studies of higher olefins are limited. Ethylene is the smallest olefin and has been studied extensively [6–10]. The most recent ethylene pyrolysis study by Ferris et al. [10] reported species concentrations using combined laser and gas chromatography (GC) techniques at 5 atm in the temperature range 1200–2000 K.

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Table 1
Cases investigated in the present study.

Case	Initial composition	Reflected pressures
A	1% C ₂ H ₄ /0.5% Kr/98.5% Ar	1 bar, 2 bar
B	2% C ₂ H ₄ /0.5% Kr/97.5% Ar	2 bar
C	2% C ₃ H ₆ /0.5% Kr/97.5% Ar	2 bar
D	2% 1-C ₄ H ₈ /0.5% Kr/97.5% Ar	2 bar
E	2% 1-C ₅ H ₁₀ /0.5% Kr/97.5% Ar	2 bar
F	2% 1-C ₆ H ₁₂ /0.5% Kr/97.5% Ar	2 bar

Propene pyrolysis was studied in a shock tube by Burcat [11], Chappell and Shah [12], Kiefer et al. [13] and Hidaka et al. [14]. Moreover, Davis et al. performed pyrolysis studies in a flow reactor [15]. However, propene pyrolysis has not recently been investigated in a shock tube and experimental data from current state-of-the-art facilities may improve our understanding of its pyrolysis.

1-Butene pyrolysis was studied by Zhang et al. [16] at 4–12 Torr in a flow-reactor coupled to a synchrotron vacuum ultraviolet (VUV) photoionization mass spectrometer using a molecular-beam sampling technique. Powers and Corcoran [17] studied 1-butene pyrolysis in the temperature range 803–876 K in a micro-reactor. More recently, Pinkowski et al. [18] carried out a multi-wavelength speciation study of 1-butene pyrolysis at 1.8 atm and 1300 K. Wang et al. [19] studied butene isomer pyrolysis in a flow reactor at 0.82 atm in the temperature range 808–1083 K. Most of the studies on 1-butene pyrolysis were conducted at atmospheric or sub-atmospheric pressures, and thus revisiting the pyrolysis of 1-butene at elevated pressures and also over a wide temperature range is valuable.

1-Pentene pyrolysis was studied by Tsang [20] over a narrow temperature range of 1000–1200 K. Tsang considered a retro-ene reaction as a new reaction class for fuel consumption together with other reaction classes from lower 1-alkenes, including unimolecular dissociation and H-atom abstraction and addition reactions.

Finally, 1-hexene was studied by Tsang [20] in the temperature range 977–1170 K using a SPST. In addition, Kiefer et al. [21] carried out a laser schlieren study in the temperature range 1220–1700 K at 50 and 200 Torr using 2% and 3% 1-hexene in Kr. The unimolecular decomposition of 1-hexene was investigated by King [22] in the temperature range 915–1153 K. The formation of H atoms during the pyrolysis of 1-hexene was studied by Peukert et al. [23]. The measurement of species profiles for 1-hexene pyrolysis at high temperatures is not present in the literature.

To provide complementary data and aid in the validation of a chemical mechanism describing the pyrolysis of 1-alkenes at high temperatures, we have studied the pyrolysis of 1% and 2% ethylene (C₂H₄), and 2% propene (C₃H₆), 1-butene (1-C₄H₈), 1-pentene (1-C₅H₁₀) and 1-hexene (1-C₆H₁₂) at 2 ± 0.16 bar in a SPST, Table 1. Furthermore, no experimental data was found for 1-olefins $\geq$ C₄ over the conditions studied.

2. Experimental work

2.1. NUIG single-pulse shock tube (SPST)

The NUIG SPST has a 5.8 m driven section with an inner diameter of 10.24 cm. There are 5 PCB Piezotronics 113A21 transducers embedded in the sidewall of the tube located at distances of 2.99, 23.69, 36.59, 49.49 and 188.72 cm from the endwall. The endwall has a Kistler 603CAB piezoelectric sensor connected to a Kistler 5018 charge amplifier. The driver section is barrel shaped with a diameter of 52 cm and a length of 53 cm. The two main components of a SPST are the tee section and the dump tank. A custom-made tee section is connected to a pneumatically actuated Festo VZBF ball valve with a response time of approximately

300 ms. This valve is connected to a dump tank. The tee section also houses a pneumatic linear actuator to burst the diaphragm. The diameter of the tee section is the same as that of the driven section. A dump tank is a vital component of a SPST. It is a pressure vessel into which the reflected shock wave is absorbed and ensures that there is no re-heating from multiple reflections of the shock wave. The dump tank assembled at NUIG is a 200 L ASME 150 class SS316 pressure vessel rated to a pressure of 15 bar. This vessel is connected to an Omega digital pressure gauge and has connections for evacuation and for argon delivery. The dump tank is connected to a Festo ball valve which was described earlier.

2.2. Experimental procedure

Our mixture preparation technique has been described by Peluchi et al. [24] and Gillespie [25]. The driver and driven sections of the tube are separated using a polycarbonate diaphragm. The ball valve to the dump tank is closed. The driven section is evacuated to $\sim 10^{-4}$ Torr using a diffusion pump and the pressure is monitored using an Edwards APG-L Pirani gauge. Once a pressure of $\sim 10^{-4}$ Torr is achieved, the driven section is charged with the fuel mixture to the desired pressure (p_1), which is monitored using a KJLC capacitance manometer. This pressure is obtained from an in-house Python program based on the shock jump equations. The dump tank is evacuated to 10^{-2} Torr using a two-stage RV pump and is filled with argon to the same pressure as the driven section. Thereafter, the driver section, which is initially evacuated, is filled with helium to the required pressure (p_4), using an Omega digital pressure gauge. For a given set of driver and driven gases at an initial temperature T_1 , the ratio of p_4 to p_1 determines the shock speed and is usually higher than the ideal ratio obtained from the shock jump equations. The valves for the diaphragm bursting mechanism, dump-tank and sampling lines are sequentially controlled using an Arduino Mega controller. Once the driven, dump-tank and driver sections are filled, the system control is transferred to the electronics system which is used to trigger the shock. The dump tank valve is opened first with a pneumatic semi-rotary actuator, and the diaphragm is burst after a delay of 800 ms with a four-pronged blade connected to the pneumatic linear actuator. Pressure traces are recorded using the pressure sensors described earlier, which are connected to a Windows PC through two Handyscope HS4s. The shock velocity and the residence time are obtained by post-processing the pressure traces with the aid of an in-house MATLAB based program. The reflected pressure and temperature conditions are then calculated using GASEQ [26], by providing the necessary inputs of initial pressure, temperature, mixture composition, and shock velocity. The calculated pressures are verified using the endwall Kistler pressure sensor.

The shock heated analyte is sampled from the endwall using a solenoid valve through a 1/16th inch tube that protrudes 8 mm into the shock tube and is analyzed using an Agilent 6890/5975 gas chromatography-mass spectrometry (GC-MS) system which is attached directly to the shock tube. The dead volume is minimal (75 mm³) due to the small diameter of the sampling tube and therefore, it is assumed that the effect of the unreacted mixture is negligible. In this study, the MS system is used to identify and quantify Kr (an internal standard) and an FID is used for all of the other organic species. The sample is introduced to a GS-Gaspro column through a split/split-less inlet which is maintained at 200 °C. Helium is used as the carrier gas at a constant flow rate of 0.9 ml min⁻¹. The temperature programming of the GC was optimized for every fuel and the system was calibrated using a 23 gas GC standard obtained from BOC Ireland. The calibrated standard provides the sensitivity of the detector for each species and this is used to calculate the concentration of the shock-heated analyte as defined by Yasunaga and Tranter [27]. The sensitivity, S ,

against Kr as an internal standard is calculated for species X in a series of calibration experiments in which measured area ratios are related to known concentration ratios as:

$$S = (A_X/A_{Kr}) * [Kr]/[X] \quad (1)$$

where A is the peak area and the square brackets denote concentration.

The concentrations of species, [X], as a result of shock heating are estimated using the sensitivity together with measured area ratios and the known concentration of Kr:

$$[X] = (1/S) * (A_X/A_{Kr}) * [Kr]. \quad (2)$$

For species with no calibration standard, the effective carbon number method was used to estimate concentrations [28]. The uncertainty in species whose peaks overlap with other species is higher than listed above. However, in the current set of experiments, we did not find any detectable co-eluting species. The shock tube was also cleaned frequently to ensure that there is no soot on the walls.

2.3. Mixtures

Ethylene ($\geq 99.5\%$ pure) and 1-butene ($\geq 99\%$ pure) were obtained from Air Liquide UK. Propene ($\geq 99.5\%$ pure), 1-pentene ($\geq 98.5\%$ pure) and 1-hexene ($\geq 99\%$ pure) were obtained from Sigma Aldrich. Pure-shield argon (Ar, $\geq 99.998\%$ pure) supplied by BOC Ireland was used as the bath gas. 99.99% pure krypton (Kr) obtained from Sigma Aldrich was used as an internal standard. 99.9% pure helium supplied by BOC Ireland was used as the driver gas. For all experiments, a mixture containing the fuel, Kr and Ar was prepared based on partial pressures measured using a KJLC capacitance manometer in a 40 L mixing vessel. The experimental procedure is provided as Supplementary Material (SM).

2.4. Experimental uncertainties

The uncertainties in reflected temperatures are calculated based on uncertainties in shock velocities of NUIG SPST and are approximately $\pm 2\%$ based on calculations by Petersen et al. [29]. The uncertainty in calibrated species concentrations calculated using repetitive sampling of the standard gas is approximately $\pm 10\%$, and estimated species concentrations, calculated using the effective carbon number method [28], is approximately $\pm 20\%$. The uncertainty in reactant mole fractions is $\pm 0.02\%$. The uncertainty in the residence time is $\pm 2\%$. The 2σ variation in calculated reflected pressures is approximately $\pm 8\%$. The carbon balances from the GC-MS measurements were $100 \pm 15\%$.

3. Kinetic modeling

Simulations were performed using Chemkin-Pro [30] assuming a closed homogeneous batch reactor at constant volume. There are typically two different approaches used to simulate a SPST: (1) based on the actual recorded pressure profiles, (2) based on the residence/reaction time. We found no significant differences [31] in the results using the two approaches as shown by Han et al. [32] for the conditions tested and presented in the SM. This is true for all of the cases and therefore, we used the residence time approach for all our simulations in the 3–4 ms range, with details provided as SM. AramcoMech3.0 was used as the reference mechanism [33]. Modifications in the rate constants of important reactions were made and are provided as NUIGMech1.0 in the SM.

Thermodynamic data for C_0 – C_3 species are primarily from Burcat's database [34]. However, where possible, both the reaction rate and thermochemical properties are adopted from recent high-accuracy theoretical studies to maintain thermodynamic consistency with the calculated rate constant. In the absence of data

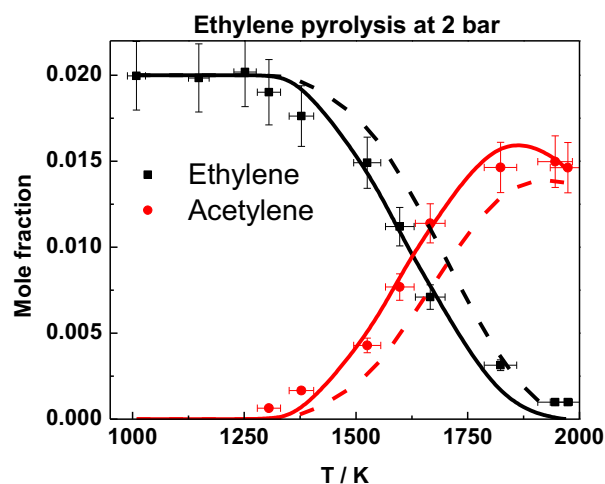


Fig. 1. Species profiles for ethylene pyrolysis. Solid lines –this study, dashed lines – AramcoMech3.0.

from reliable literature or database, the group additivity method has been used to calculate the thermodynamic parameters using the THERM code for all other species. We compare and evaluate thermodynamic data when available before including them in the mechanism, and the results using the THERM code are generally very similar to quantum chemistry or experimental results. We frequently update the group additivity values used in THERM from the literature [35–37].

4. Comparison with experimental results

4.1. Ethylene

Ethylene (2%) experiments were carried out at 2 bar in the temperature range 1000–1950 K. Acetylene (C_2H_2) is the only major product, Fig. 1. Vinyl acetylene and diacetylene were found to be formed in concentrations below 100 ppm (the quantification threshold for the current GC-MS system).

Ethylene is converted to vinyl radicals via H-atom abstraction by $\dot{H}$ atoms and vinyl radicals decompose to acetylene and $\dot{H}$ atoms. The improvement in ethylene predictions is due to the modification to the H-atom abstraction rate constant from ethylene by $\dot{H}$ atoms. This rate constant has been computed recently at NUIG, and agrees well with the rate constant data from Shao et al. [38]. The model predicts the formation of vinyl acetylene and di-acetylene at higher temperatures.

Furthermore, we compared our 1% ethylene pyrolysis experiments at 1 bar and 2 bar to those performed by Ferris et al. [10] at 5 atm. The pressure dependence observed in the experimental results is well captured by the model as shown in Fig. 2, and lends supporting evidence to the pressure dependent reaction rates for ethylene pyrolysis.

4.2. Propene

Propene experiments were performed at 2 bar in the temperature range 1000–1800 K. The major products are methane, ethylene, allene (C_3H_4 -a), propyne (C_3H_4 -p) and acetylene, with minor products including ethane (C_2H_6), 1,3-butadiene (C_4H_6), vinyl acetylene (C_4H_4), diacetylene (C_4H_2) and benzene (C_6H_6), Fig. 3.

A reaction path analysis for propene pyrolysis at 1450 K is provided in Fig. 4.

$\dot{H}$ atom assisted reactions, either via abstraction or addition reactions are the main propene consumption pathways. $\dot{H}$ atom addition and subsequent decomposition of propyl radicals, to produce

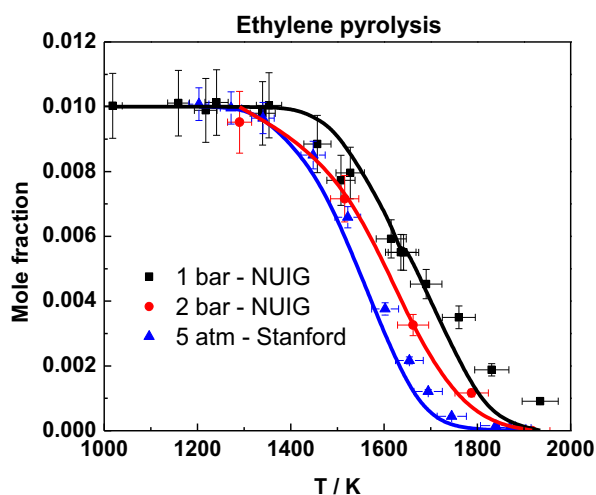


Fig. 2. Ethylene consumption as a function of temperature at 1 bar, 2 bar and 5 atm.

ethylene and methyl radicals is the major consumption pathway of propene. Abstraction of allylic hydrogen atoms by $\dot{\text{H}}$ atoms and methyl ($\dot{\text{C}}\text{H}_3$) radicals to produce allyl radicals and H_2 and methane (CH_4) contribute approximately 38% to propene consumption. The unimolecular initiation reaction of propene, to produce allyl radicals and $\dot{\text{H}}$ atoms contribute only about 6.9% to propene's consumption. Through the various abstraction and unimolecular initiation reactions approximately 45% of propene is converted to allyl radicals, which primarily decompose to allene. This subsequently isomerizes to propyne or undergoes H-atom abstraction reactions to produce propargyl radicals. Consumption of propyne was found to be dominated by reactions with $\dot{\text{H}}$ atoms and methyl radicals. Reactions of propyne with $\dot{\text{H}}$ atoms predominantly produce acetylene and methyl radicals, with only small amounts of flux observed to proceed through H-atom abstraction reactions to produce propargyl radicals. On the other hand, reactions of propyne with methyl radicals primarily produce propargyl radicals, which produces benzene isomers (benzene, fulvene, 2-ethynyl-1,3-butadiene).

Acetylene is formed by the decomposition of vinyl radicals, the reaction of $\dot{\text{H}}$ atoms with allene and propyne, and β -scission of propen-1-yl radical.

It is interesting to note that we do not see quantifiable amounts of 1,5-hexadiene (<100 ppm), which is expected to be produced from the recombination of allyl radicals. This can be attributed to the fall-off characteristics of the reaction with temperature, and

also could be that high temperatures facilitate $\dot{\text{H}}$ atom elimination reactions producing allene.

4.3. 1-Butene

1-Butene experiments were performed at 2 bar in the range of 900–1700 K. The major products are methane, ethylene, allene, propyne, 1,3-butadiene and acetylene. Minor products include vinyl acetylene, 1,2-butadiene, 1,5-hexadiene and cyclopentadiene. The measured species profiles along with the predictions from the current mechanism and AramcoMech 3.0 are presented in Fig. 5.

The current mechanism captures the decay of the parent fuel and the evolution of most of the intermediate species more accurately than AramcoMech3.0 with major differences observed only in the case of mole fractions of vinyl-acetylene, Fig. 5. The rate constants for unimolecular fuel decomposition reactions are the same in NUIGMech1.0 and AramcoMech3.0, but the rate constants for 1-butene + $\dot{\text{H}}$ and 1-butene + $\dot{\text{C}}\text{H}_3$ were updated based on those calculated by Zhang et al. [39] and Wang et al. [40] respectively. The current mechanism uses rate constants from Li et al. for 1,3-butadiene + $\dot{\text{H}}$ [41].

Additionally, the disproportionation reaction of 1-buten-3-yl and allyl radicals to form propene and 1,3-butadiene was updated using a rate rule analogy from the allyl-allyl disproportionation reaction forming allene and propene [42]. The main improvement in the predictions using NUIGMech1.0 stems from the change in choice of rate constant for H-atom abstraction from the fuel by methyl radicals and the subsequent reactions of the 1-buten-3-yl ($\dot{\text{C}}_4\text{H}_7\text{-1-3}$) radical formed. Furthermore, the updated chemically activated pathways significantly improved the ethylene and propene predictions.

Figure 6 presents a reaction flux diagram for 1-butene pyrolysis at 1234 K. The pyrolysis chemistry of 1-butene is very similar to that of propene with $\dot{\text{H}}$ atom assisted reactions (abstraction and addition), and abstraction by methyl radicals being the important pathways. $\dot{\text{H}}$ atom addition reactions to 1-butene produce propene and methyl radicals and ethylene and ethyl ($\dot{\text{C}}_2\text{H}_5$) radicals via two separate chemically activated pathways. Ethyl radicals decompose to produce ethylene and $\dot{\text{H}}$ atoms. H-atom abstraction from 1-butene by $\dot{\text{H}}$ atoms and $\dot{\text{C}}\text{H}_3$ radicals leads to the formation of 1-buten-3-yl ($\dot{\text{C}}_4\text{H}_7\text{-1-3}$) radicals which in turn produce 1,3-butadiene via β -scission. One striking difference in the chemistry of propene and 1-butene is in the relative importance of the unimolecular initiation reaction. The unimolecular initiation reaction in 1-butene (and higher 1-olefins) is more facile than for propene due the presence of a weak allylic C–C bond (BDE ~ 71 kcal/mole). The unimolecular initiation reaction produces allyl and methyl radicals. Methyl radicals subsequently participate in a variety of reactions such as (i) self-recombination reactions to produce ethane;

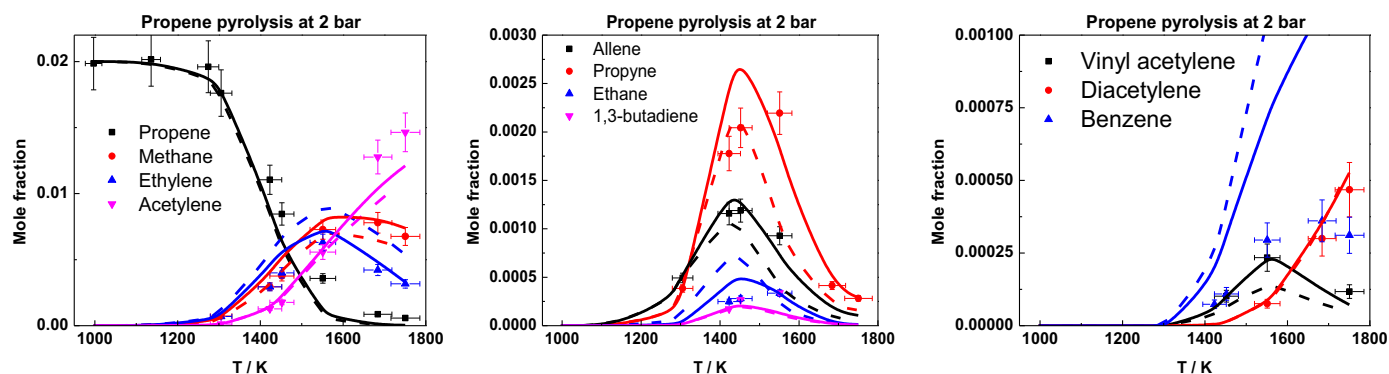


Fig. 3. Species profile for propene pyrolysis. Solid lines – NUIGMech1.0, dashed lines – AramcoMech3.0.

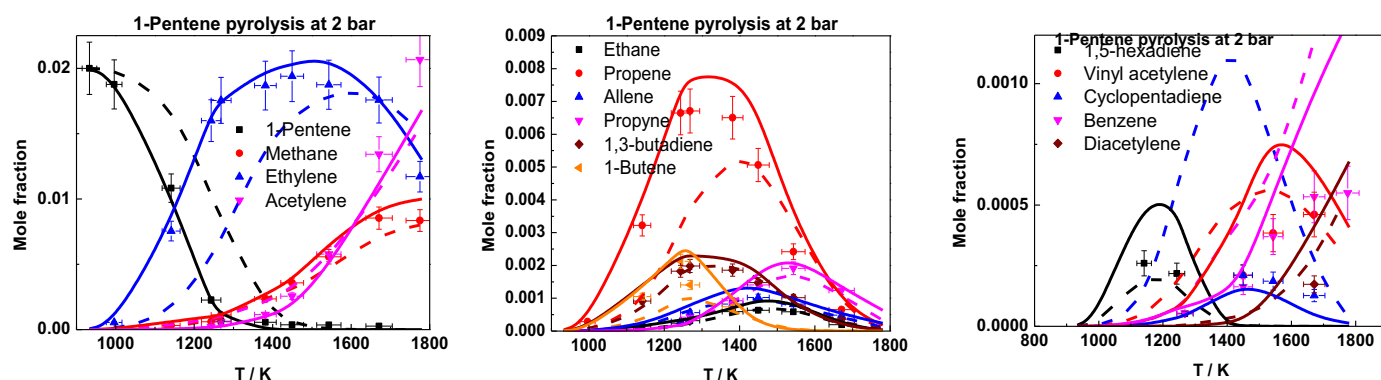


Fig. 7. Species profiles for 1-pentene pyrolysis. Solid lines – NUIGMech1.0, dashed lines – AramcoMech3.0.

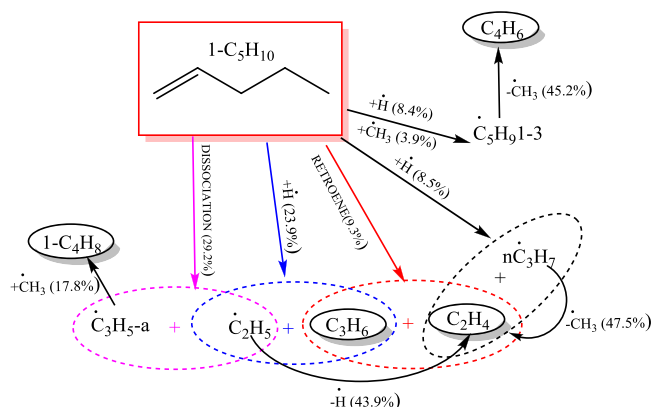


Fig. 8. Reaction flux diagram of 1-pentene pyrolysis at ~20% fuel consumption, 1141 K, 2.144 bar.

major products are methane, ethylene, allene, propyne, propene, 1,3-butadiene and acetylene. Minor products include ethane, vinyl acetylene, 1-butene, benzene, 1,5-hexadiene, diacetylene and cyclopentadiene. AramcoMech3.0 does not contain a complete set of C_5 species and hence some of these profiles are not presented for its predictions.

The unimolecular dissociation reaction rate constant is updated and the retro-ene reaction is added from Tsang [43] in NUIGMech1.0 compared to AramcoMech3.0. The 1-pentene + $\dot{H}$, 1-pentene + $\dot{CH}_3$ and 1,3-pentadiene + $\dot{H}$ rate constants are updated based on calculations by Power et al. [35], Wang et al. [40] and Sun et al. [44] respectively. The main differences between the two mechanism originates from the updated reactions, mostly stemming from the retro-ene reaction ($1-C_5H_{10} \rightarrow C_2H_4 + C_3H_6$).

A reaction pathway analysis of 1-pentene pyrolysis at 1141 K is illustrated in Fig. 8. The important reaction pathways in the pyrolytic chemistry of 1-pentene are similar to that of 1-butene, with the exception of the retro-ene reaction. This reaction leads to the production of propene and ethylene from 1-pentene. The $\dot{H}$ atom addition reactions of 1-pentene produce propene and ethyl radicals or ethylene and propyl radicals via chemically activated pathways.

H-atom abstraction from 1-pentene produces 1-penten-3-yl ($\dot{C}_5H_9-1-3$) radicals which dissociate into 1,3-butadiene and methyl radicals. The dissociation (29%) of 1-pentene produces allyl radicals which decompose to form allene and/or its self-disproportionation reaction produces allene and propene. Allyl radicals can also undergo a recombination reaction with itself or methyl radicals to produce 1,5-hexadiene and 1-butene, respectively. Methyl radical recombination and disproportionation reactions of allyl and ethyl radicals are the main sources of ethane. Propyne is produced via

the isomerization of allene. The production of methane and acetylene occurs via pathways similar to those discussed above for 1-butene.

We also detect considerable amounts of cyclopentadiene and benzene during pyrolysis of 1-pentene. Cyclopentadiene (C_5H_6) is produced via 1,4-pentadien-3-yl (CVCCJVC) radicals which are formed from linear pentadienes. CVCCJVC can isomerize to form a cyclopentenyl radical ($\dot{C}_5H_7$) or cyclopentadiene plus $\dot{H}$ atoms. Also, allyl reacts with acetylene to form C_5H_6 and H atoms. Benzene is produced from cyclopentadiene and from the recombination of propargyl radicals through pathways shown in our earlier study [45].

4.5. 1-Hexene

1-Hexene experiments were performed at 2 bar in the range of 900–1700 K. The major products are methane, propene, ethane, ethylene, acetylene, allene, propyne, propene, 1-butene and 1,3-butadiene, Fig. 9. Minor products include 1-pentene, vinyl acetylene, benzene, 1,5-hexadiene, diacetylene and cyclopentadiene. A new mechanism for C_6 species based on the LLNL gasoline surrogate mechanism [46] was formulated and is included in NUIGMech1.0.

A reaction flux analysis for 1-hexene decomposition at 1131 K is shown in Fig. 10. Unimolecular fuel decomposition occurs via C–C bond cleavage producing allyl and n -propyl radicals. 1-Butene is formed primarily via methyl addition to allyl radicals. The retro-ene reaction of 1-hexene produces two propene molecules.

Also, propene is formed via the chemically activated pathway, $C_6H_{12} + \dot{H} = C_3H_6 + n\dot{C}_3H_7$. Ethylene is formed by multiple channels, with the major pathways from n -propyl and ethyl radical β -scission reactions. 1,3-butadiene is formed via the decomposition of 1-hexen-3-yl ($\dot{C}_6H_{11}-1-3$) and 1-butenyl ($\dot{C}_4H_7-1-3$ and $\dot{C}_4H_7-1-4$) radicals. Allene and H atoms are formed from the decomposition of allyl radicals and propyne is formed via the isomerization of allene. Methane and acetylene formation pathways are similar to those for 1-butene pyrolysis. Ethane is formed primarily via methyl radical recombination. Disproportionation reactions of allyl and ethyl radicals and H-atom abstraction by ethyl radicals from stable species also contribute to ethane formation.

The mechanism under-predicts the formation of 1,3-butadiene, with the major pathways from the dissociation of 1-butenyl and 1-hexenyl radicals. From our previous results we found that the 1-butenyl channel predictions are sufficiently accurate. Therefore, we believe that this under-prediction originates from other major channels such as the recombination reaction between methyl and propargyl radicals. This recombination reaction in the current mechanism is known to have uncertainties as large as a factor of five [47]. So, this could be the other potential reason for the

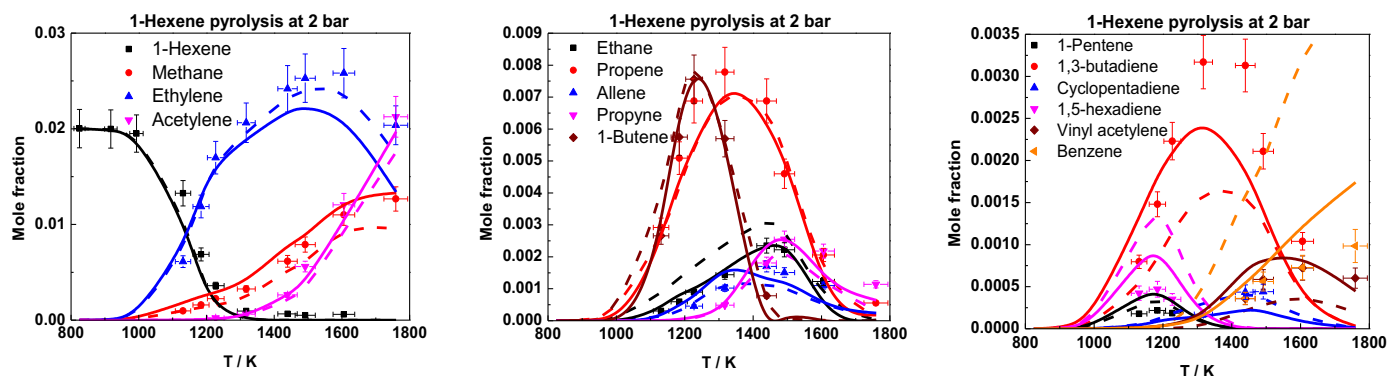


Fig. 9. Species profiles for 1-hexene pyrolysis. Solid lines – NUIGMech1.0, dashed lines – LLNL gasoline surrogate mechanism [46].

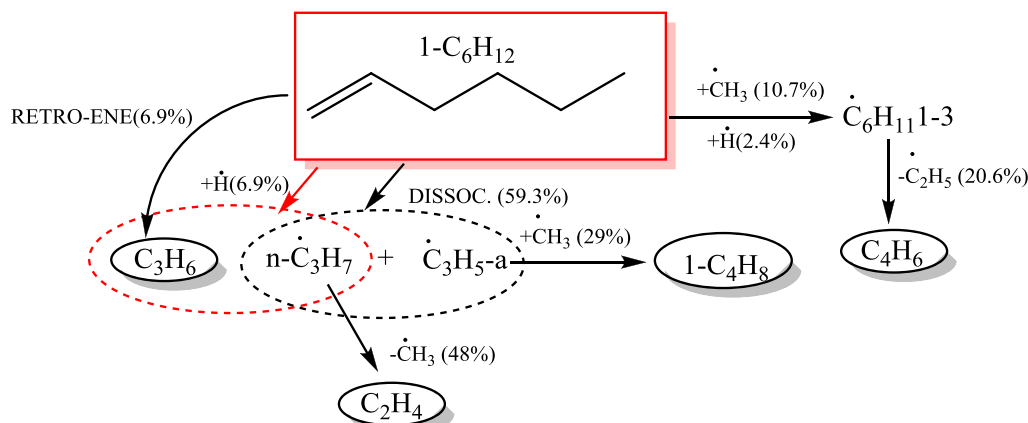


Fig. 10. Reaction flux diagram for 1-hexene pyrolysis at 2.10 bar, 1130.8 K, 20% fuel consumption.

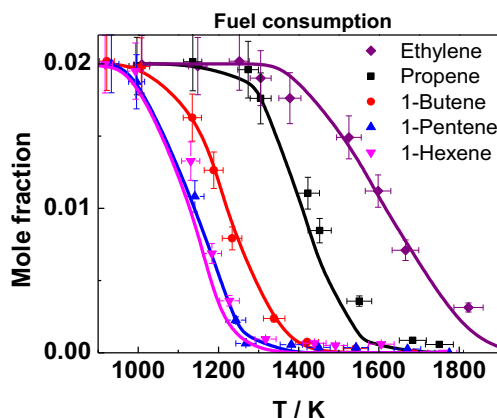


Fig. 11. Fuel mole fraction profiles for the pyrolysis of 1-alkenes at 2 bar. Solid symbols are experimental results and solid lines are simulation results using NUIG-Mech1.0.

discrepancy in butadiene concentrations, and further investigation is needed for this reaction. The mechanism is currently under-predicting the peak concentration of 1,3-butadiene by ~20%.

5. Reactivity trends of 1-olefins

With this series of experiments, we observed that the reactivity of 1-alkenes increases with increasing carbon number but the margin of increase between them reduces for carbon numbers larger than four, Fig. 11. To investigate this further, an integrated reaction pathway analysis is performed at temperatures where ~50% of

fuel is consumed. This aids in offsetting the effect of temperature and helps in objectively analyzing the reaction pathways, Fig. 12.

For ethylene pyrolysis, fuel consumption via unimolecular dissociation reactions is less prominent. Due to the presence of only vinylic hydrogen atoms and double bonded carbon atoms, the reactivity of ethylene is significantly lower than for all of the other 1-olefins. The rate constants for unimolecular dissociation used in NUIGMech1.0 are plotted in Fig. 13(a). Even though the rate constants for the unimolecular decomposition of 1-pentene and 1-hexene are almost identical, Fig. 13(a), their rates of consumption are different, as shown in the rate of production (ROP) analysis, Fig. 13(b). This is mainly due to the products formed during unimolecular decomposition. Odd carbon species ultimately produce a smaller olefin and H atoms, while even number ones produce a smaller olefin and a methyl radical. The chemically activated pathways which involve H atoms compete with unimolecular dissociation in consuming the fuel and this is more prominent for odd carbon-number fuels due to the higher concentration of available H atoms, Fig. 13(c). However, 1-butene has a high contribution from chemically activated pathways and is an exception.

Therefore, for fuel consumption, the contribution of unimolecular decomposition is highest (~64%) for 1-hexene and it decreases to approximately 30% for 1-pentene, 24% for 1-butene, 10% for propene and 5% for ethylene. This sequence is true at different temperatures and pressures (with details provided as SM). Furthermore, chemically activated reactions of the fuel with hydrogen atoms play a vital role in consuming the fuel for shorter alkenes and its significance gradually decreases with increasing carbon number for even carbon number fuels. H-atom abstraction by $\dot{\text{C}}\text{H}_3$ radicals is dominant for propene, 1-butene and 1-hexene consumption. However, this plays a lesser role in 1-pentene consumption

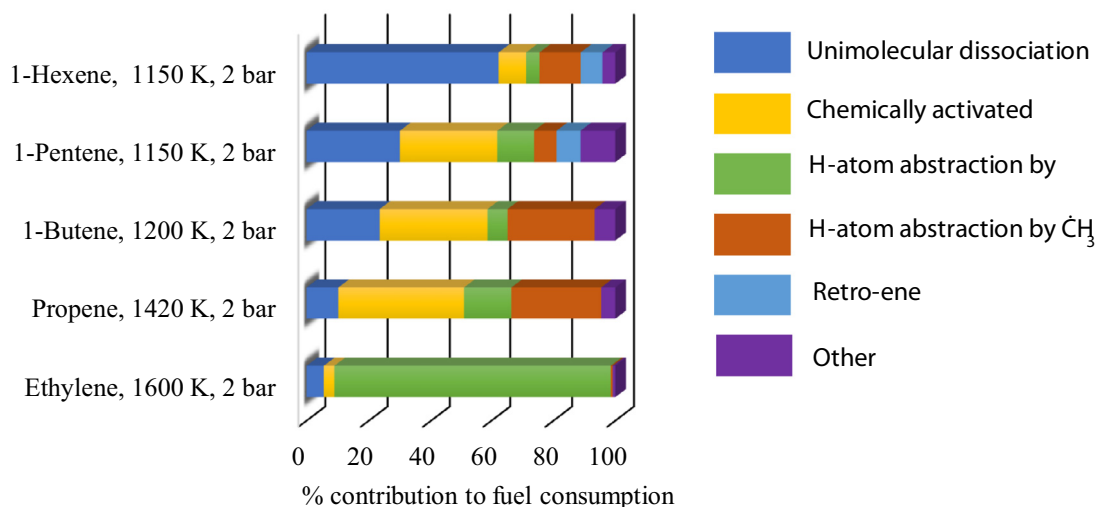
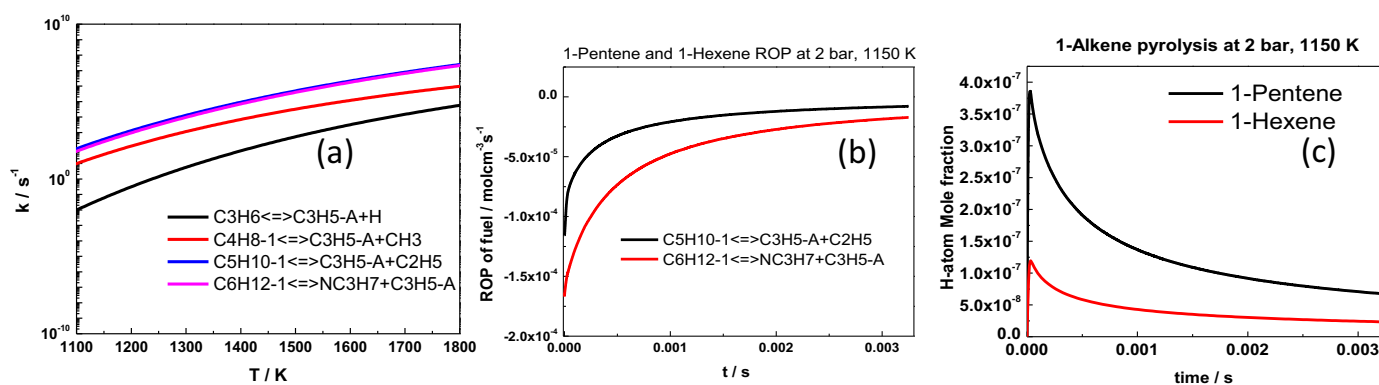
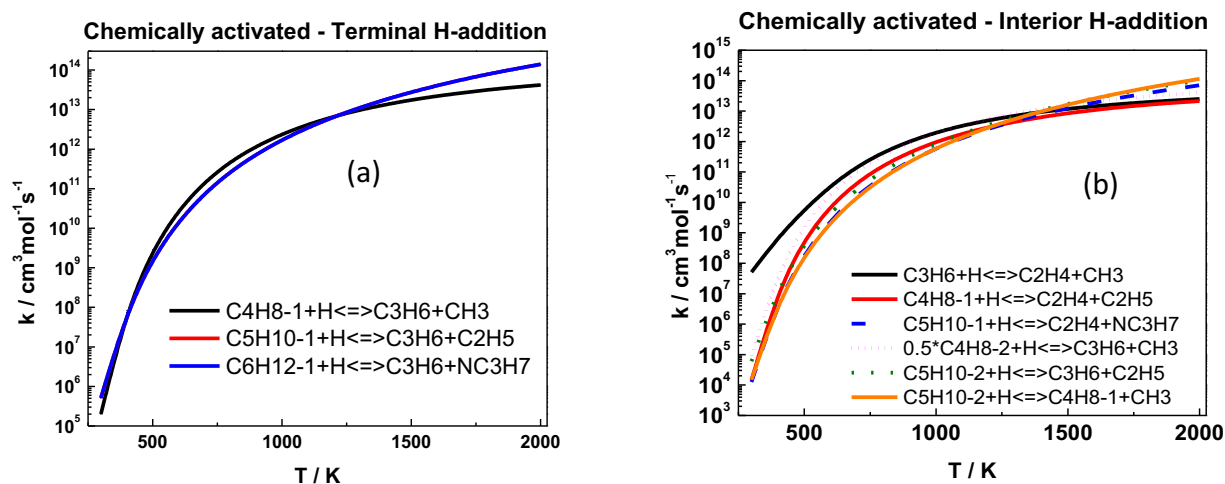


Fig. 12. Integrated reaction pathway analysis for 1-alkene fuel consumption.

Fig. 13. (a) Unimolecular dissociation rate constants used in the mechanism; (b) Unimolecular dissociation rates; (c) $\dot{\text{H}}$ atom mole fraction for 1-pentene and 1-hexene at 2 bar and 1150 K.Fig. 14. Rate constants for chemically activated pathways for olefinic species by (a) terminal $\dot{\text{H}}$ addition; (b) interior $\dot{\text{H}}$ addition.

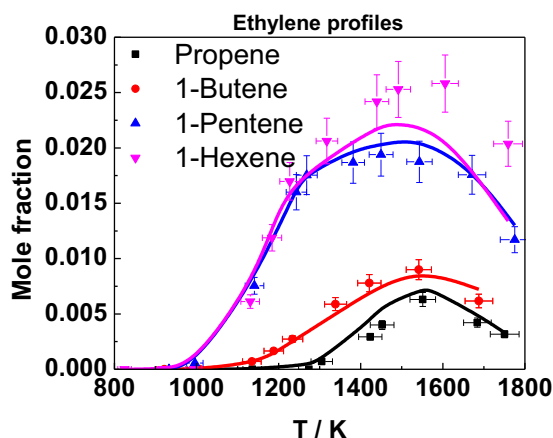


Fig. 15. Ethylene mole fraction from 1-alkene pyrolysis.

due to the lower concentration of methyl radicals produced by unimolecular dissociation or chemically activated pathways. Additionally, retro-ene reactions consume fuels for 1-alkenes $\geq C_5$.

After illustrating the importance of chemically activated fuel + $\dot{H}$ and retro-ene reactions, their high-pressure rate constants are compared. The current mechanism uses theoretically calculated propene + $\dot{H}$ reactions from Miller and Klippenstein [48], 1-butene, 2-butene + $\dot{H}$ reactions from Li et al. [49,50] and 1-pentene, 2-

pentene + $\dot{H}$ reactions from Power et al. [35]. The rate constants from linear pentenes + $\dot{H}$ are used for linear hexenes + $\dot{H}$. The different rate constants for chemically activated pathways for terminal $\dot{H}$ addition for 1-alkenes at vinylic position and internal $\dot{H}$ addition for 1- and 2-alkenes at vinylic position are shown in Fig. 14(a) and (b).

Rate constants used for retro-ene reactions of 1- $C_5H_{10} \rightarrow C_2H_4 + C_3H_6$ and 1- $C_6H_{12} \rightarrow C_3H_6 + C_3H_6$ are from the studies by Tsang [20,43]. These rates are very consistent with each other partially due to the assumption a similar A-factor for 1-pentene and could be used for higher 1-olefins using the rate rule analogy.

Ethylene concentrations increase with increasing fuel carbon number at a given temperature, as shown in Fig. 15 and there is a significant increase in ethylene when the carbon number increases from four (1-butene) to five (1-pentene), but there is a moderate change in ethylene concentrations when it increases from five (1-pentene) to six (1-hexene). These differences are primarily linked to decomposition pathways of fuel as illustrated in Fig. 16.

Reactions (R3) and (R4) show that the decomposition of C_5 and C_6 1-olefins produce ethylene which doesn't occur for C_3 and C_4 1-olefins, explaining the jump in C_2H_4 concentration as the size of olefin increases from 1-butene to 1-pentene. Furthermore, the odd carbon number species produce $\dot{H}$ atoms upon decomposition while even carbon number species produce $\dot{CH}_3$ radicals, explaining the similarity in the concentrations of methane produced for odd and even carbon number fuels, Fig. 17(a). To corroborate this,

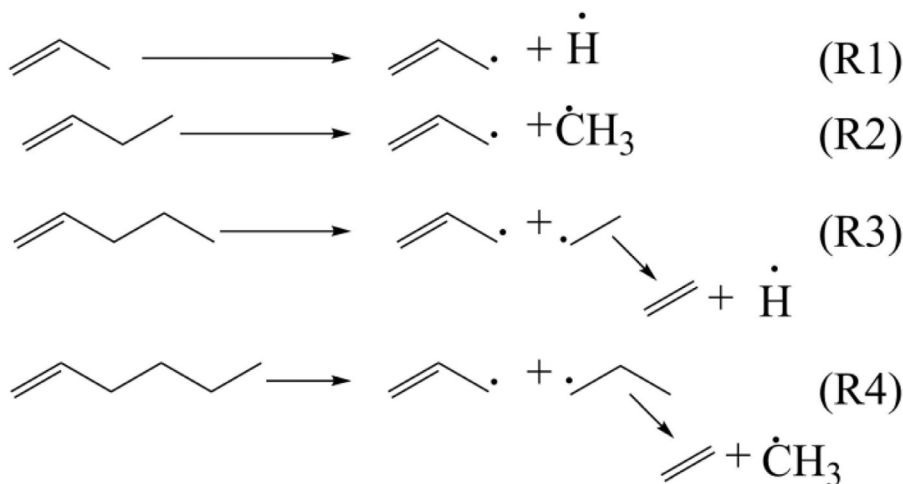


Fig. 16. Unimolecular decomposition pathways of 1-alkenes.

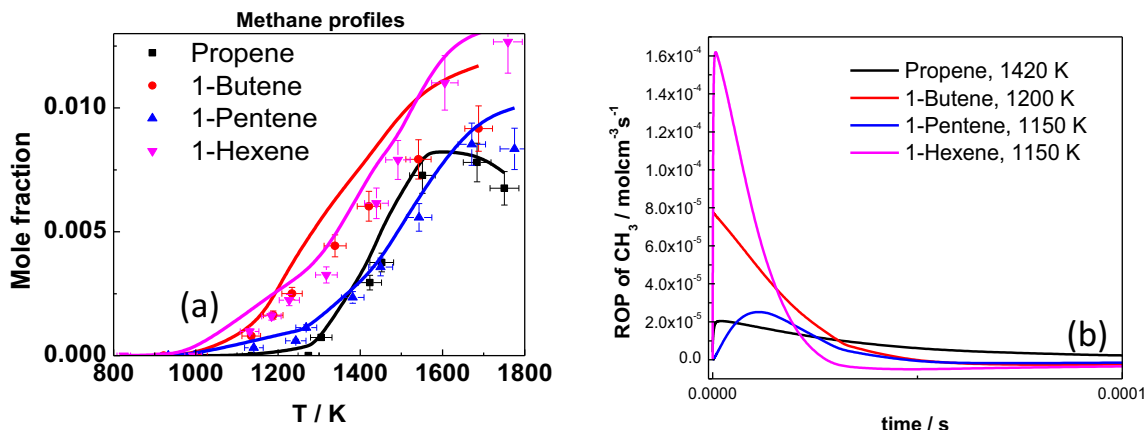


Fig. 17. (a) Methane concentration profiles for 1-alkenes pyrolysis; (b) ROP analysis for $\dot{CH}_3$ radical at ~50% fuel consumption of 1-alkenes.

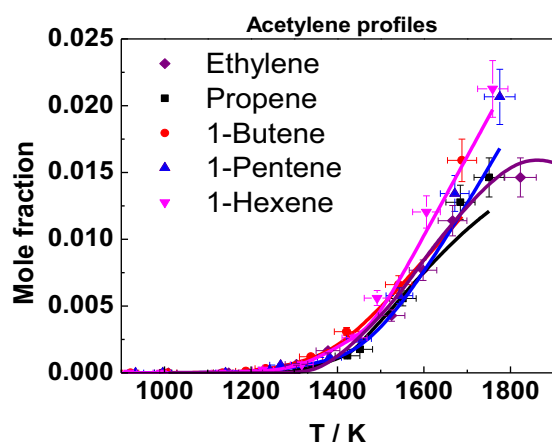


Fig. 18. Acetylene formation from pyrolysis of 1-alkenes.

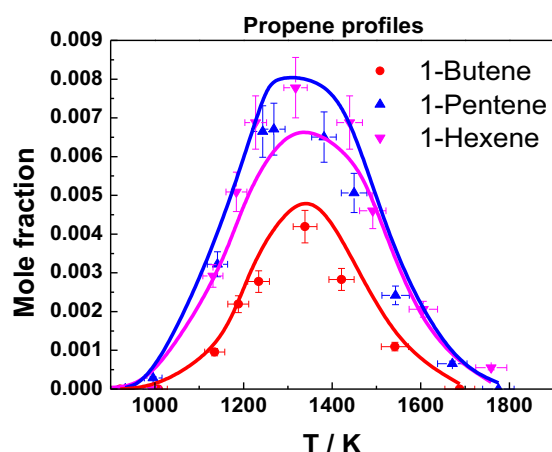


Fig. 19. Propene profiles from 1-alkenes pyrolysis.

a ROP analysis of CH_3 radicals, using NUIGMech1.0, shows that methyl radical production is higher for fuels with an even number of carbon atoms, Fig. 17(b).

The formation of acetylene starts at 1400 K for all fuels, Fig. 18, and differences in acetylene concentrations are seen at higher temperatures. The formation of acetylene is controlled by vinyl radicals produced from consumption of ethylene and as seen in Figs. 1 and 15, the decay of ethylene starts at $T = 1400$ K.

Finally, propene is formed from 1-butene, 1-pentene and 1-hexene mainly via the chemically activated pathway of fuel + H producing propene and a methyl, ethyl or *n*-propyl radical respectively. The increased concentration in the cases of 1-pentene and 1-hexene, Fig. 19 is due to the presence of the retro-ene reaction, $1\text{-C}_5\text{H}_{10} \rightarrow \text{C}_2\text{H}_4 + \text{C}_3\text{H}_6$ and $1\text{-C}_6\text{H}_{12} \rightarrow \text{C}_3\text{H}_6 + \text{C}_3\text{H}_6$. This establishes the need to include such reactions when developing mechanisms for higher hydrocarbons.

This current study provides new experimental data for mechanism validation and also provides an insight about how the different reaction classes play a role in fuel consumption at pyrolytic conditions. Furthermore, it highlights the role of unimolecular dissociation in the formation of ethylene and explains the differences in methane concentrations for the odd and even carbon number 1-olefins. This can be extended for higher hydrocarbons. Additionally, it is seen that acetylene is mainly formed from ethylene decomposition for all of the fuels studied. The presence of a retro-ene reaction is also illustrated by comparing the propene profiles for 1-butene, 1-pentene and 1-hexene. Overall, this study provides

new experimental data and tries to comprehend the results using a hierarchical chemical kinetics approach.

6. Conclusions

The pyrolysis of ethylene, propene, 1-butene, 1-pentene and 1-hexene was studied in a SPST at a reflected pressure of 2 ± 0.16 bar and over a temperature range of 900–1800 K. The mole fractions of C_1 – C_6 species produced during pyrolysis of the 1-olefins were identified and quantified using a GC–MS and an FID respectively. A new detailed kinetic mechanism, NUIGMech1.0, used in our simulations was found to satisfactorily predict the effect of chain length on the production of C_1 , C_2 and C_3 fragments, and predicts the concentration of all the major species detected. The speciation results from these experiments contribute to a database for the validation of mechanisms for pyrolytic conditions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2020.06.021](https://doi.org/10.1016/j.combustflame.2020.06.021).

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